All Sections | Time Left: 179:27 | Question Paper | Instructions

Question No. 1 #665df48dca324ed4a4c50fbe1 | Section Details | Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

A body starts from rest and moves with uniform acceleration. If the distance travelled by it in the first 2 s is x_1 and in the next 2 s is x_2, then x_1 and x_2 are related as

- $x_1 = x_2$
- $x_1 = 2x_2$
- $2x_1 = x_2$
- $3x_1 = x_2$

18 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT

Choose a question

1	2	3	4	5	6
7	8	9	10	11	12
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19	20				

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All Sections | Time Left: 179:20 | Question Paper | Instructions

Question No. 2 #665df48dca324ed4a4d58fbe2 | Section Details | Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Statement (A): When the temperature increases the viscosity of gases increases and the viscosity of liquids decreases.

Statement (B): Water does not wet an oily glass because cohesive force of oil is less than that of water.

Statement (C) : A liquid will wet a surface of a solid if the angle of contact is greater than 90°.

- A. B. and C are false
- A and B false. C is true
- B and C false. A is true
- A and C false. B is true

18 Not Visited | 0 Marked for Review

0 Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT

Choose a question

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All Sections | Time Left: 179:16 | Question Paper | Instructions

Question No. 3 #665df48dca324ed4a4d5f0be3 | Section Details | Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The refractive index of a prism for a monochromatic wave is $\sqrt{2}$ and its refracting angle is 60°. For minimum deviation, the angle of incidence will be

- 30°
- 45°
- 60°
- 75°

17 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT

Choose a question

1	2	3	4	5	6
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All Sections | Time Left: 179:11 | Question Paper | Instructions

Question No. 4

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Statement (I) : A uniform electric field and a uniform magnetic field are pointed in the same direction. If an electron is projected in the same direction the electron velocity will decrease in magnitude.
Statement (II) : Two infinite long parallel wires are carrying equal current in the same direction. The magnetic field at a point midway between the wires is zero.
Statement (III) : No net force acts on a rectangular coil carrying a steady current when suspended in a uniform magnetic field.
Which of the following is correct?

- Statements I, II and III are true.
- Statements I and II are true, but statement III is false.
- Statements II and III are true, but statement I is false.
- Statements I and III are true, but statement II is false.

Mark for Review & Next | Clear Response | Save & Next | Submit

Review
Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

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All Sections | Time Left: 179:7 | Question Paper | Instructions

Question No. 5 #665df48dca324ed4a4d5bfbe5 | Section Details | Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The escape velocity for the earth is Ves. Find out escape velocity for a planet whose radius is four times and density is nine times that of the earth. If is $n \times 6$ Ves. Find n.

- 6
- 4
- 8
- 2

15 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT

Choose a question

1	2	3	4	5	6
7	8	9	10	11	12
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19	20				

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All Sections | Time Left: 179:3 | Question Paper | Instructions

Question No. 6 #665df48dca324ed4a4c50fbe6 | Section Details | Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

A railway track is banked for a speed v, by making the height of the outer rail h higher than that of the inner rail. If the distance between the rails is l and the radius of curvature of the track is r, then

- $\frac{h}{l} = \frac{v^2}{rg}$
- $\tan\left\{\sin^{-1}\left(\frac{h}{l}\right)\right\} = \frac{v^2}{rg}$
- $\tan^{-1}\left(\frac{h}{l}\right) = \frac{v^2}{rg}$
- $\frac{h}{r} = \frac{v^2}{lg}$

14 Not Visited | 0 Marked for Review
0 Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT

Choose a question

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20				

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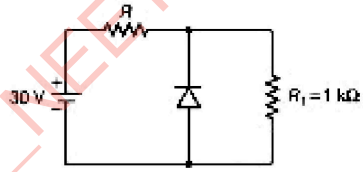
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All Sections | Time Left: 178:59 | Question Paper | Instructions

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

If current in diode is five times that in R_1 and breakdown voltage of diode is 6 V, find the value of R ?



- ○ 2000Ω
- ○ $\frac{2000}{3}\ \Omega$
- ○ 1000Ω
- ○ $\frac{1000}{3}\ \Omega$

Mark for Review & Next | Clear Response | Save & Next | Submit

0 Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT

Choose a question

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All Sections | Time Left: 178:56 | Question Paper | Instructions

Only One Option Correct Type

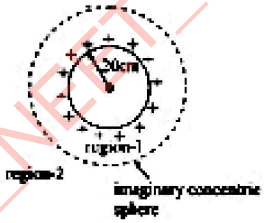
This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

PHYSICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

What is the radius of the imaginary concentric sphere that divides the electrostatic field of a metal sphere of a radius 20 cm and a charge of 8μC in two regions of identical energy. If it is n × 10 cm then find n.



- 1
- 2
- 3
- 4

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All Sections | Time Left: 178:51 | Question Paper | Instructions

Question No. 9 #665df48dca324ed4a4c50fbe9 | Section Details | Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Which of the following statements is wrong?

- KE of a body is independent of the direction of motion
- In an elastic collision of two bodies ,the momentum and energy of system is conserved
- If two protons are brought towards each other the PE of the system increases.
- A body cannot have energy without momentum.

Mark for Review & Next | Clear Response | Save & Next | Submit

11 Not Visited | 0 Marked for Review

0 Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT

Choose a question

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13	14	15	16	17	18
19	20				

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All Sections | Time Left: 178:42 | Question Paper | Instructions

Question No. 10 #665df48dca324ed4a4c90fbe10 | Section Details | Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

A gas in container A is in thermal equilibrium with another gas in container B. Both contain equal masses of the two gases in the respective containers. Which of the following can be true?

- ○ $P_A V_A = P_B V_B$
- ○ $P_A = P_B, V_A \neq V_B$
- ○ $P_A = P_B, V_A = V_B$
- ○ $\frac{P_A}{V_A} = \frac{P_B}{V_B}$

Mark for Review & Next | Clear Response | Save & Next | Submit

10 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

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All Sections | Time Left: 178:37 | Question Paper | Instructions

Question No. 11 #665df48dca324ed4a4c98be11 | Section Details | Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Let A_n be the area enclosed by the nth orbit in a hydrogen atom. The graph of $\ell n(A_n/A_1)$ against ℓ n(n)

- will not pass through the origin
- will have certain points lying on a straight line with slope 4
- will be a monotonically increasing non linear curve
- will be a circle

Mark for Review & Next | Clear Response | Save & Next | Submit

8 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

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All Sections | Time Left: 178:29 | Question Paper | Instructions

Only One Option Correct Type

be considered for evaluation)

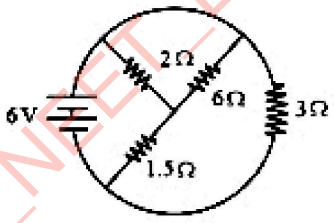
This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

PHYSICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

The total current supplied to the circuit by the battery is



- 1 A
- 2 A
- 4 A
- 6 A

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All Sections | Time Left: 178:22 | Question Paper | Instructions

Question No. 13 #665df48dca324ed4a4c50dbe13 | Section Details | Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

On six complete rotations, the screw gauge moves by 3 mm on the main scale. If there are 50 divisions on the circular scale the least count of the screw gauge is

- 0.01 cm
- 0.02 mm
- 0.001 mm
- 0.001 cm

7 Not Visited | 0 Marked for Review

0 Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

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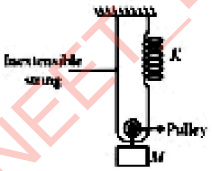
This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

PHYSICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

The time period of mass M when displaced from its equilibrium position and then released for the system as shown in figure is



- $2\pi\sqrt{\frac{M}{k}}$
- $2\pi\sqrt{\frac{M}{2k}}$
- $2\pi\sqrt{\frac{M}{4k}}$
- $2\pi\sqrt{\frac{2M}{k}}$

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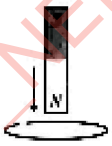
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Question No. 15 #6605bf4bbca324ec4a4f50fbe15 | Section Details | Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

A copper ring having a cut such as not to form a complete loop is held horizontally and a bar magnet is dropped through the ring with its length along the axis of the ring, figure. The acceleration of the falling magnet is



- g
- Less than g
- More than g
- Zero

Mark for Review & Next | Clear Response | Save & Next | Submit

5 Not Visited | 0 Review | 0 Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
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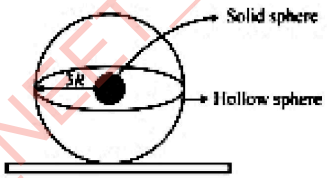
This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

PHYSICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

A solid ball of mass m and radius R is released from the position shown in a large hollow fixed shell of same mass m and radius $3R$ as shown in the figure. The displacement of the centre of mass of the system from its initial position when the solid ball touches the lower surface of the hollow shell is (centres of both the spheres coincide initially):



- 0
- $3R$
- $2R$
- R

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All Sections | Time Left: 178:4 | Question Paper | Instructions

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

A plane electromagnetic wave travelling along the x-direction has a wavelength of 3 mm. The variation in the electric field occurs in the y-direction with an amplitude 66 $\mathrm{Vm^{-1}}$. The equation for the electric and magnetic fields as a function of x and t are respectively

- $E_y = 33\cos\pi \times 10^{11}\left(t - \frac{x}{c}\right)$ and $B_z = 1.1 \times 10^{-7}\cos\pi \times 10^{11}\left(t - \frac{x}{c}\right)$
- $E_y = 11\cos 2\pi \times 10^{11}\left(t - \frac{x}{c}\right)$ and $B_y = 11 \times 10^{-7}\cos 2\pi \times 10^{11}\left(t - \frac{x}{c}\right)$
- $E_z = 33\cos\pi \times 10^{11}\left(t - \frac{x}{c}\right)$ and $B_y = 11 \times 10^{-7}\cos\pi \times 10^{11}\left(t - \frac{x}{c}\right)$
- $E_y = 66\cos 2\pi \times 10^{11}\left(t - \frac{x}{c}\right)$ and $B_z = 2.2 \times 10^{-7}\cos 2\pi \times 10^{11}\left(t - \frac{x}{c}\right)$

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All Sections | Time Left: 178:0 | Question Paper | Instructions

Question No. 18 | Section Details | Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Which of these is correct match?

A.	Angular momentum	1.	$\left[M^{-1}L^{2}T^{-2}\right]$
B.	Torque	2.	$\left[M^{1}T^{-2}\right]$
C.	Gravitational constant	3.	$\left[M^{1}L^{2}T^{-2}\right]$
D.	Tension	4.	$\left[M^{1}L^{2}T^{-1}\right]$

- ○ C – 2, D – 1
- ○ A – 4, B – 3
- ○ A – 3, C – 2
- ○ B – 2, A – 1

Mark for Review & Next | Clear Response | Save & Next | Submit

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2 Not Visited | 0 Review
0 Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT
Choose a question
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13 14 15 16 17 18
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All Sections | Time Left: 177:57 | Question Paper | Instructions

Question No. 19 #665df48dca324ed4a4c90fbe19 | Section Details | Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

n identical resistance are taken in which $\frac{n}{2}$ resistors are joined in series in the left gap and the remaining $\frac{n}{2}$ resistances are joined in parallel in the right gap of a metre bridge. Balancing length in cm is

- $100 \cdot \frac{n^2}{n^2+4}$
- $100 \cdot \frac{n^2}{n^2+1}$
- $400 \cdot \frac{1}{n^2+4}$
- $400 \cdot \frac{1}{n^2+1}$

1 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

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All Sections | Time Left: 177:54 | Question Paper | Instructions

Question No. 20 #665df48dca324ed4f4e50fbe20 | Section Details | Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Assertion (A) When an ideal gas is compressed adiabatically, its temperature and the average kinetic energy of the gas molecules increase.
Reason (R) The kinetic energy increases because of collisions of molecules with moving parts of wall only.

- A and R and true and R is correct explanation of A.
- A and R are true but R is not correct explanation of A.
- A is true and R is false.
- A is false and R is true.

Mark for Review & Next | Clear Response | Save & Next | Submit

0 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT

Choose a question

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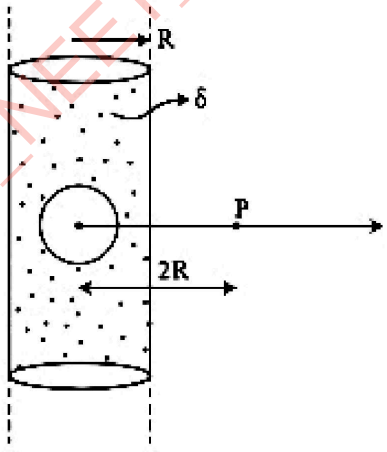
This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

PHYSICS NUMERICAL

Choose a question

1 2 3 4 5

An infinitely long solid cylinder of radius R with uniform volume charge density ρ has a spherical cavity of radius $\frac{R}{2}$ with its centre on the axis of the cylinder as shown in the figure. The magnitude of the electric field at a point P which is at a distance $2R$ from the axis of the cylinder is given by $\frac{23\rho R}{6K\epsilon_0}$. What is the value of K?



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This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

PHYSICS NUMERICAL

Choose a question

1 2 3 4 5

Calculate the energy released in MeV in the following nuclear reaction :

$^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He + Q$

[Mass of $^{238}_{92}U$ = 238.05079 u]

Mass of $^{234}_{90}Th$ = 234.043630 u

Mass of $^{4}_{2}He$ = 4.002600 u

1 u = 931.5 MeV (mark answer to nearest integer in MeV)

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7 8 9

4 5 6

1 2 3

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All Sections | Time Left: 177:40 | Question Paper | Instructions

Physics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

Three identical uniform thin metal rods form the three sides of an equilateral triangle. If the moment of inertia of the system of these three rods about an axis passing through the centroid of the triangle and perpendicular to the plane of the triangle is n times the moment of inertia of one rod separately about an axis passing through the centre of the rod and perpendicular to its length, the value of n is

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4 5 6
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Clear All

Answered and Marked for Review (will be considered for evaluation)

PHYSICS NUMERICAL

Choose a question

1 2 3 4 5

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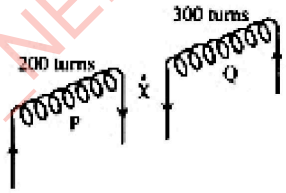
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All Sections | Time Left: 177:37 | Question Paper | Instructions

Physics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

Two solenoids P and Q of equal length but different number of turns, are arranged coaxially as shown in figure. P has 200 turns and Q has 300 turns. There is a current of 1 A in Q. the current in P in order that there is no resultant field at X midway between the coils is K, find 8 K.



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7 8 9
4 5 6

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0 Answered and Marked for Review (will be considered for evaluation)

PHYSICS NUMERICAL

Choose a question

1 2 3 4 5

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All Sections | Time Left: 177:33 | Question Paper | Instructions

Question No. 5 #665df48dca324ed4a4c50fbe5 | Section Details | Marking Scheme

0 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

Physics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

Bulk modulus of water is 2×10^9 N/m^2. The pressure required to increase the volume of water by 0.1% in N/m^2 is 2×10^x N/m^2. Find K.

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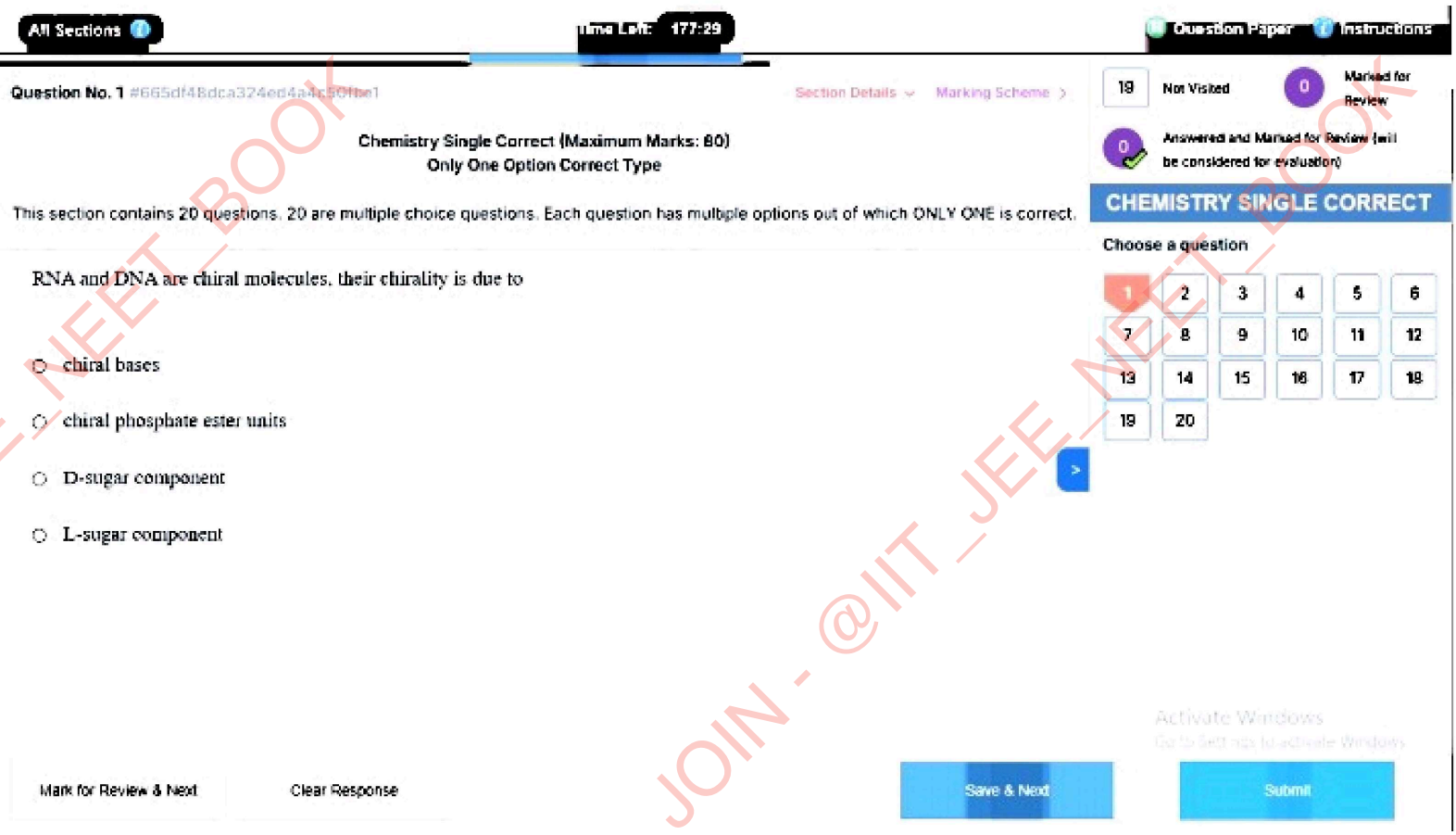
PHYSICS NUMERICAL
Choose a question
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Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 177:26 | Question Paper | Instructions

Question No. 2 #665df48dca324ed4a4d58fbe2 | Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

$Mg(OH)_2$ is precipitated when NaOH is added to a solution of Mg^{2+}. If the final concentration of Mg^{2+} is 10^{-10} M, the concentration of OH^- (M) in the solution is

[Solubility product for $Mg(OH)_2 = 5.6 \times 10^{-12}$]

- ○ 0.056
- ○ 0.12
- ○ 0.24
- ○ 0.025

18 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

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Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 177:20 | Question Paper | Instructions

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

CHEMISTRY SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

Match the following.

List I (Element)	List II (Property)
A. He	I. Can be used as inert atmosphere
B. Ar	II. Radioactive element
C. Xe	III. Form fluoride compounds
D. Rn	IV. Can be used as airship balloons

The correct match is
A B C D

- IV III II I
- IV I III II

Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 177:16 | Question Paper | Instructions

Only One Option Correct Type

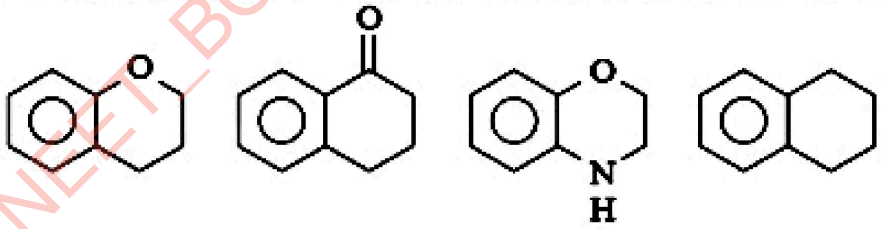
This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

CHEMISTRY SINGLE CORRECT

Choose a question

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20				

Rank the following compounds in decreasing order of reactivity in electrophilic aromatic substitution reaction



(1) (2) (3) (4)

- ○ 3 > 1 > 2 > 4
- ○ 4 > 3 > 2 > 1
- ○ 3 > 1 > 4 > 2
- ○ 1 > 3 > 4 > 2

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Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 177:12 | Question Paper | Instructions

Question No. 5 #665df48dca324ed4a4d58fbe5 | Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

In an atom the order of increasing energy of electrons with quantum numbers
(i) $n = 4, \ell = 1$
(ii) $n = 4, \ell = 0$
(iii) $n = 3, \ell = 2$ and (iv) $n = 3, \ell = 1$ is

- iii < i < iv < ii
- ii < iv < i < iii
- i < iii < ii < iv
- iv < ii < iii < i

Mark for Review & Next | Clear Response | Save & Next | Submit

15 Not Visited | 0 Marked for Review
0 Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

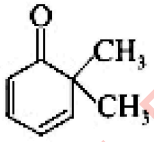
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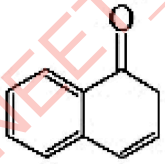
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19	20				

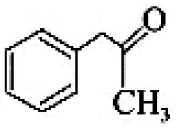
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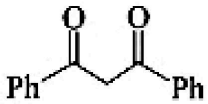
All Sections | Time Left: 176:59 | Question Paper | Instructions

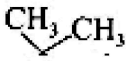
Which of the following carbonyl compounds will exhibit enolization?

(i) 

(ii) 

(iii) 

(iv) 



1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

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Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 176:55 | Question Paper | Instructions

Question No. 7 #665df48dca324ed4a4c5fbe7 | Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

For BCl_3, $AlCl_3$ and $GaCl_3$ the increasing order of ionic character is

- $BCl_3 < AlCl_3 < GaCl_3$
- $GaCl_3 < AlCl_3 < BCl_3$
- $BCl_3 < GaCl_3 < AlCl_3$
- $AlCl_3 < BCl_3 < GaCl_3$

13 Not Visited | 0 Marked for Review

0 Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 176:52 | Question Paper | Instructions

Question No. 8 #665df48dca324ed4a4d58fbe8 | Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The boiling point of an azeotropic mixture of water and ethyl alcohol is less than that of theoretical value of water and alcohol mixture. Hence, the mixture shows

- that solution is highly saturated
- positive deviation from Raoult's law
- negative deviation from Raoult's law
- nothing can be said

12 Not Visited | 0 Marked for Review

0 Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
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All Sections | Time Left: 176:49 | Question Paper | Instructions

Question No. 9 #665df48dca324ed4a4d58fbe9 | Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Which complex among the following has the highest value of spin only magnetic moment?

$[Fe(CN)_6]^{3-}$, $[Fe(CN)_6]^{4-}$, $[Ni(CN)_4]^{2-}$, $[NiCl_4]^{2-}$

- $[Fe(CN)_6]^{3-}$
- $[Fe(CN)_6]^{4-}$
- $[Ni(CN)_4]^{2-}$
- $[NiCl_4]^{2-}$

11 Not Visited | 0 Marked for Review

0 Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

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All Sections | Time Left: 176:46 | Question Paper | Instructions

Question No. 10 #665df48dca324ed4a4c50fbe10 | Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Directions : Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.
Assertion: o-Hydroxy benzene sulphonic acid on heating to 373 K changes to p-Hydroxybenzene sulphonic acid.
Reason: Sulphonation of phenol is a reversible process.

- If both assertion and reason are true and reason is the correct explanation of assertion.
- If both assertion and reason are true but reason is not the correct explanation of assertion.
- If Assertion is true but reason is false.
- If both assertion and reason are false.

Mark for Review & Next | Clear Response | Save & Next | Submit

10 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

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All Sections | Time Left: 176:39 | Question Paper | Instructions

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

Consider the following configurations:



Which of the above configuration may represent the excited state of C-atom.

- (i), (iii) & (iv)
- (i) only
- (i) & (iii)
- (iii) only

Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 176:36 | Question Paper | Instructions

Question No. 12 #665df48dca324ed4a4c50fbe12 | Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Assertion (A) Boron has a smaller first ionisation enthalpy than beryllium.
Reason (R) The penetration of a 2*s*-electron to the nucleus is more than the 2*p*-electron hence, 2*p*-electron is more shielded by the inner core of electrons than 2 s-electrons.

- Assertion and Reason both are correct statements but Reason is not correct explanation for Assertion.
- Assertion is correct statement but Reason is wrong statement.
- Assertion and Reason both are correct statements and Reason is correct explanation for Assertion.
- Assertion and Reason both are wrong statements.

Mark for Review & Next | Clear Response | Save & Next | Submit

8 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

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All Sections | Time Left: 176:31 | Question Paper | Instructions

Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

CHEMISTRY SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

The order of strengths of the following carboxylic acids is

(i) $CH_3 - CH_2 - COOH$

(ii) $CH_3 - COOH$

(iii) $C_6H_5 - COOH$

(iv) $C_6H_5 - CH_2COOH$

- (iii) > (iv) > (ii) > (i)
- (iv) > (ii) > (iii) > (i)
- (iii) > (ii) > (iv) > (i)
- (i) > (iv) > (ii) > (iii)

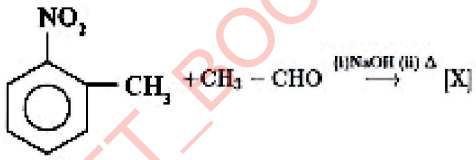
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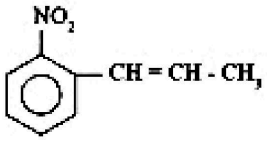
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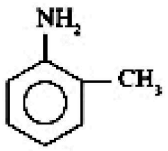
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In the given reaction



Then [X] will be





1 2 3 4 5 6
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Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 176:18 | Question Paper | Instructions

Question No. 15 #665df48dca324ed4a4c50dbe15 | Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Which of the following is not correct about Grignard reagent?

- It is a nucleophile
- Forms new carbon-carbon bond
- Reacts with carbonyl compounds
- It is an organomanganese compound

5 Not Visited | 0 Marked for Review

0 Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
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All Sections | Time Left: 176:16 | Question Paper | Instructions

Question No. 16 #665df48dca324ed4a4c90fbe16 | Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The strongest Lewis acid is -

- BF_3
- BCl_3
- BBr_3
- BI_3

4 Not Visited | 0 Marked for Review

0 Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
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All Sections | Time Left: 176:12 | Question Paper | Instructions

Question No. 17 #665df48dca324ed4a4c50fbe17 | Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

A compound A contains C and H only and has molecular mass 72. It's photochlorination gives a mixture containing only one monochloro and two dichloro hydrocarbons. The IUPAC name is

- 2, 2-dimethylpropane
- n-pentane
- 2-methyl butane
- cyclopentane

3 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

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Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 176:8 | Question Paper | Instructions

Question No. 18 #665df48dca324ed4a4c90fbe18 | Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Which of the following statement(s) is (are) correct when a mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4? Mark incorrect option

- A deep red vapour is evolved
- The vapour when passed through NaOH solution, gives a yellow solution
- Chlorine gas is also evolved
- Chromyl chloride is formed

Mark for Review & Next | Clear Response | Save & Next | Submit

2 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

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7 8 9 10 11 12
13 14 15 16 17 18
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All Sections | Time Left: 176:6 | Question Paper | Instructions

Question No. 19 #665df48dca324ed4a4c50dbe19 | Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

2 –Bromobutane (A) is treated with NaI in the presence of dry acetone to give compound 'B'. The compound 'B' is boiled with moist silver oxide to give compound 'C'. Identify the compounds 'B' and 'C'. How many optical isomers are possible for compound 'C'?

- 2 –Iodobutane, Butan–2 –ol, 2 isomers
- 3 –Iodopentane, Pentan–3 –ol, 4 isomers
- 2 –Iodobutane, Butan–2 –ol, 4 isomers
- 3 –Iodopentane, Pentan–3 –ol, 2 isomers

1 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 176:1 | Question Paper | Instructions

Question No. 20 #665df48dca324ed4a4c50fbe20 | Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Choose the correct option regarding the following.
Assertion : Actinides exhibit a larger number of oxidation states than lanthanides.
Reason : There is a large energy gap between $5f$, $6d$ and $7s$ sub-shells.

- Assertion and Reason are correct, and Reason is the correct explanation for the Assertion.
- Assertion and Reason are correct, and Reason is not the correct explanation for the Assertion.
- Assertion is correct, Reason is wrong.
- Assertion is incorrect, Reason is correct.

Mark for Review & Next | Clear Response | Save & Next | Submit

0 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

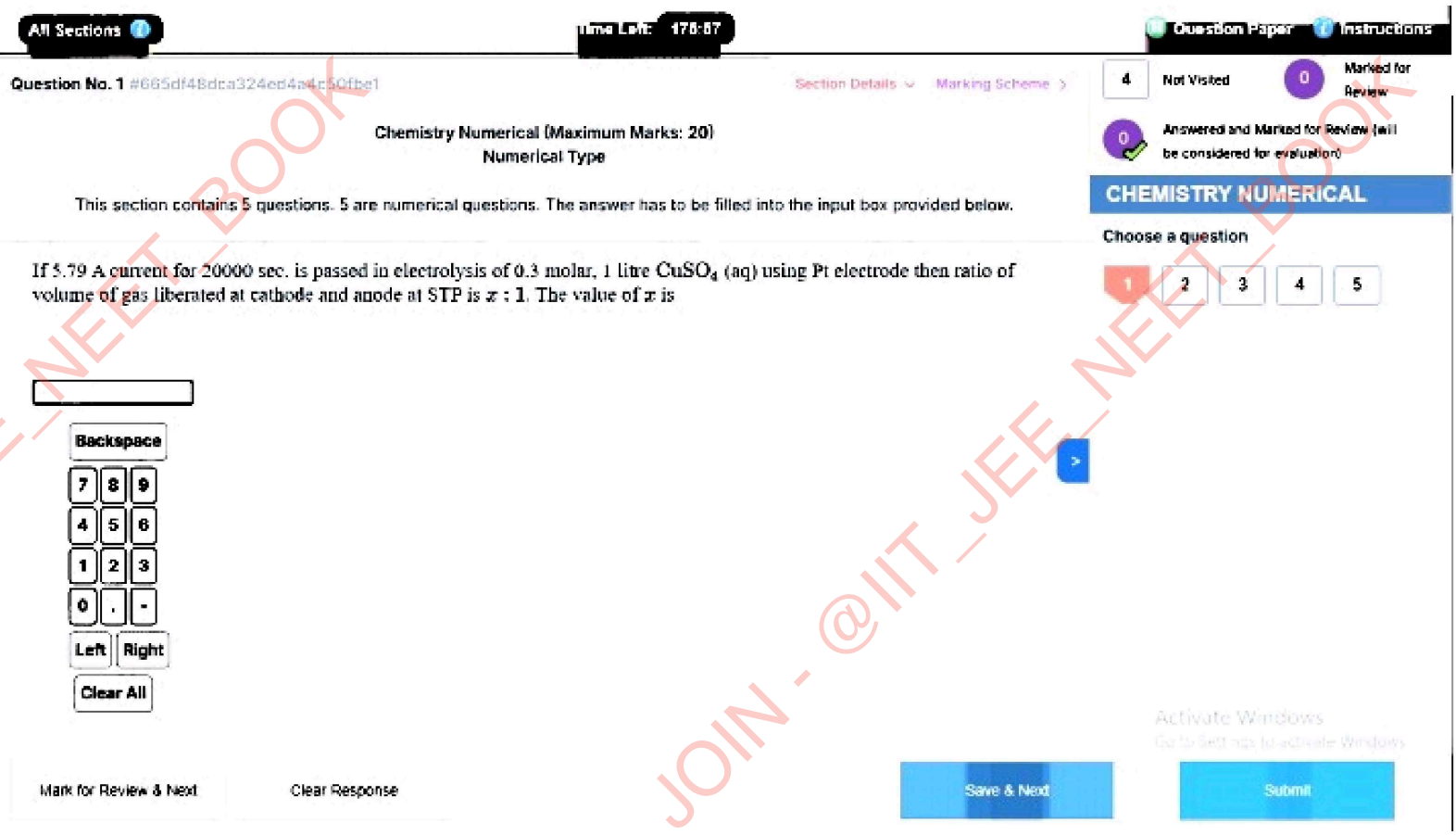
Choose a question

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All Sections | Time Left: 178:52 | Question Paper | Instructions

Question No. 2 #665df48dca324ed4a4c50fbe2 | Section Details | Marking Scheme

Chemistry Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

The concentration of R in the reaction $R \rightarrow P$ was measured as a function of time and the following data is obtained:

[R](molar)	1.0	0.75	0.40	0.10
t(min).	0.0	0.05	0.12	0.18

The order of the reaction is

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Mark for Review & Next | Clear Response | Save & Next | Submit

3 Not Visited | 0 Marked for Review
0 Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY NUMERICAL

Choose a question

1 2 3 4 5

Activate Windows
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All Sections | Time Left: 178:47 | Question Paper | Instructions

Question No. 3 #665af48eca324ed4a4c50fbe3 | Section Details | Marking Scheme

Chemistry Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

During the Kjeldahl method, 2.8 g of an organic compound was digested and the evolved ammonia was absorbed by 60 mL of $\frac{M}{20}$ H_2SO_4. What volume of $\frac{M}{20}$ NaOH solution was required for complete neutralisation of the unreacted acid if the percentage of nitrogen in the compound was 1% ? (mark answer in mL)

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Mark for Review & Next | Clear Response | Save & Next | Submit

2 Not Visited | 0 Marked for Review
0 Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY NUMERICAL
Choose a question
1 2 3 4 5

Activate Windows
Go to Settings to activate Windows.
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All Sections | Time Left: 178:43 | Question Paper | Instructions

Question No. 4 #665df48dca324ed4a4c50fbe4 | Section Details | Marking Scheme

Chemistry Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

A metal (M) forms two oxides. The ratio M : O (by weight) in the two oxides are 25 : 4 and 25 : 6. The minimum value of atomic mass of M is

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Mark for Review & Next | Clear Response | Save & Next | Submit

1 Not Visited | 0 Marked for Review
0 Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY NUMERICAL
Choose a question
1 2 3 4 5

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All Sections | Time Left: 178:40 | Question Paper | Instructions

Question No. 5 #665df48dca324ed4a4cb0fbe5 | Section Details | Marking Scheme

Chemistry Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

Two mole of an ideal gas is isothermally compressed in two stages starting from (1 atm, 20 lit) to 10 lit. at external pressure P_1 then upto 2 lit at external pressure P_2. Find work done (lit-atm) in the process.

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4 5 6
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Clear All

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0 Not Visited | 0 Marked for Review
0 Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY NUMERICAL

Choose a question

1 2 3 4 5

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All Sections | Time Left: 178:36 | Question Paper | Instructions

Question No. 1 #665df48dca324ed4a4c50fbe1 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

If L and M are respectively the coefficient of x^{-7} in $\left(ax+\frac{b}{x^2}\right)^{11}$ and the coefficient of x^7 in $\left(bx^2+\frac{a}{x}\right)^{11}$ then $L+M=$

- $\frac{1}{b}\left[\text{coefficient of } x^{-6} \text{ in } \left(ax+\frac{b}{x^2}\right)^{12}\right]$
- $\frac{1}{a}\left[\text{coefficient of } x^{6} \text{ in } \left(ax^2+\frac{b}{x}\right)^{12}\right]$
- $a\left[\text{coefficient of } x^{-10} \text{ in } \left(ax+\frac{b}{x^2}\right)^{11}\right]$
- $b\left[\text{coefficient of } x^{4} \text{ in } \left(ax^2+\frac{b}{x}\right)^{11}\right]$

Mark for Review & Next | Clear Response | Save & Next | Submit

19 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20				

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All Sections | Time Left: 178:27 | Question Paper | Instructions

Question No. 2 #665df48dca324ed4a4d58fbe2 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

If the foot of the perpendicular from the point $A(p+1,-1,11)$ on the line $\frac{x}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ is $B(q,5,7)$ then the value of $(p-q)$ is ...

- 1.5
- 1
- 0.5
- 0

18 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 178:23 | Question Paper | Instructions

Question No. 3 #665df48dca324ed4a4dc50fbe3 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The particular solution of the different equation
$\frac{dx}{dy} = \frac{\sin y(1+x\cot y)}{x\log(x^2e)}, y(1) = 0$

- $y\sin y = x^2\log x$
- $y^2\sin y = \log x$
- $y = \left(\frac{e^2}{\sin 1}\right)(x-1)$
- $y = e^2\sec x$

17 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20				

Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 175:20 | Question Paper | Instructions

Question No. 4 #665df48dca324ed4a4d5f0ee4 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Let $J = \int_0^{e-1} \frac{1}{x+1} \exp\cdot\left(\frac{x^2+2x-1}{2}\right)dx$ and $K = \int_1^e x \ln x \exp\cdot\left(\frac{x^2-2}{2}\right)dx$. The value of $(J+K)$ is equal to (Here exp. (x) means e power x)

- $(\sqrt{e})^{e^2+1}$
- $(\sqrt{e})^{e^2-1}$
- 0
- $(\sqrt{e})^{e^2-2}$

16 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 178:17 | Question Paper | Instructions

Question No. 5 #665df48dca324ed4a4d58fbe5 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

If sum of coefficients in the expression of $\left(\alpha x^2 - 2x + 1\right)^{35}$ is equal to sum of coefficients in the expansion of $(x - \alpha y)^{35}$, then α is equal to

- 0
- 1
- any real number.
- 2

15 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20				

Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 178:14 | Question Paper | Instructions

Question No. 6 #665df48dca324ed4a4c58fbe6 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The number of common terms between the sequences given by $1, 4, 7, 10 \ldots\ldots 298$ and $2, 4, 6, 8 \ldots\ldots 300$ is

- 49
- 50
- 48
- 100

14 Not Visited | 0 Marked for Review

0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20				

Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 178:11 | Question Paper | Instructions

Question No. 7 #665df48dca324ed4a4c5fbe7 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The axis of a parabola lie along the line $y = x$ and the distance of its vertex from origin is $\sqrt{2}$ and that of focus is $2\sqrt{2}$. If both focus and vertex lie in the first quadrant, then the equation of the parabola will be

- $(x+y)^2 - (x-y-2)$
- $(x-y)^2 = (x+y-2)$
- $(x-y)^2 - 4(x+y-2)$
- $(x-y)^2 = 8(x+y-2)$

13 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 175:8 | Question Paper | Instructions

Question No. 8 #665df48dca324ed4a4d58fbe8 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The reciprocal of the distance between two points, one on each of the lines $\frac{x-2}{3} = \frac{y-4}{2} = \frac{z-5}{5}$ and $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$,

Then which of the following is FALSE.

- can be 8
- can be less than $\sqrt{75}$
- cannot be greater than 78
- can be $\sqrt{80}$

12 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

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All Sections | Time Left: 176:6 | Question Paper | Instructions

Question No. 9 #665df48dca324ed4a4dc50fbe9 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Let $f(x)$ be an even function, $I_1 = \int_0^{\frac{\pi}{2}} f(\cos 2x)\cos x dx$, $I_2 = \int_0^{\frac{\pi}{4}} f(\sin 2x)\cos x dx$. Then $\frac{I_1}{I_2} =$

- 1
- $\frac{1}{2}$
- $\frac{1}{\sqrt{2}}$
- $\sqrt{2}$

11 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

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All Sections | Time Left: 176:3 | Question Paper | Instructions

Question No. 10 #665df48dca324ed4a4c90fbe10 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

A relation R is defined as $(x, y) \in R \Rightarrow x^y = y^x$ for $x, y \in I - \{0\}$, where I is the set of all integers. Then the relation R is:

- reflexive but not symmetric
- symmetric but not reflexive
- reflexive and symmetric both
- equivalence relation

10 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

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All Sections | Time Left: 176:0 | Question Paper | Instructions

Question No. 11 #665df48dca324ed4a4c98fbe11 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

If z is a complex number satisfying the equation $|z-(1+i)|^2=2$ and $\omega=\frac{2}{z}$, then the locus traced by 'ω' in the complex plane is

- $x-y-1=0$
- $x+y-1=0$
- $x-y+1=0$
- $x+y+1=0$

Mark for Review & Next | Clear Response | Save & Next | Submit

9 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
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All Sections | Time Left: 174:56 | Question Paper | Instructions

Question No. 12 #665df48dca324ed4a4c50dbe12 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The distinct points $A(0,\ 0)$, $B(0,\ 1)$, $C(1,\ 0)$ and $D(2\alpha,\ 3\alpha)$ are concyclic, then

- α' can attain only rational values
- 'α' is irrational
- Cannot be concyclic for any 'α'
- None of the above

8 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 174:53 | Question Paper | Instructions

Question No. 13 #665df48dca324ed4a4c90fbe13 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Let $f(x)$ be defined by $f(x) = \begin{cases} \frac{3x-[x]}{x}, & x < 1 \\ \frac{3\cdot x^2+x}{[x]}, & x \geq 1 \end{cases}$

(where [.] is GIF) Then which is incorrect

- $f(x)$ is continuous at $x = 1$
- $f(x)$ is not differentiable at $x = 1$
- $f'\left(\frac{3}{2}\right) = -2$
- $f'(x)$ is continuous at $x = 1$

Mark for Review & Next | Clear Response | Save & Next | Submit

7 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

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All Sections | Time Left: 174:50 | Question Paper | Instructions

Question No. 14 #665df48dca324ed4a4c50fbe14 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The mean and standard deviation of 10 observations $x_1, x_2, x_3 \ldots x_{10}$ are $\bar{x}$ and σ respectively. Let 10 is added to $x_1, x_2 \ldots x_9$ and 90 is subtracted from x_{10}. If still, the standard deviation is the same, then $x_{10} - \bar{x}$ is equal to

- 35
- 45
- 55
- 50

6 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 174:47 | Question Paper | Instructions

Question No. 15 #665df48dca324ed4a4c50dbe15 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The longest distance of the point $(a, 0)$ from the curve $2x^2 + y^2 = 2x$ is

- $1 + a$
- $|1 - a|$
- $\sqrt{1 - 2a + 2a^2}$
- $\sqrt{1 - 2a + 3a^2}$

5 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 174:44 | Question Paper | Instructions

Question No. 16 #665df48dca324ed4a4c90fbe16 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Let $f(x) = \cos^2 x; g(x) = \int_0^{1/x} e^{t^2}\,dt$. The value of $\lim_{x \to 0}(f(x))^{g(x)}$ is

- 1
- e
- e^{-1}
- 0

4 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 174:41 | Question Paper | Instructions

Question No. 17 #665df48dca324ed4a4c50fbe17 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Let $\vec{a} = 2\hat{i} + \hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c} = 4\hat{i} - 3\hat{j} + 7\hat{k}$. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$, then the value of $\vec{r} \cdot \vec{b}$ is

- 7
- −7
- −5
- 5

3 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 174:38 | Question Paper | Instructions

Question No. 18 #665df48dca324ed4a4c90fbe18 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}, a, b, \in N$. Then :

- there exist more than one but finite number of B's such that $AB = BA$
- there exists exactly one B such that $AB = BA$
- there exists infinitely many B 's such that $AB = BA$
- there cannot exist any B such that $AB = BA$ C

2 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

Mark for Review & Next | Clear Response | Save & Next | Submit

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All Sections | Time Left: 174:36 | Question Paper | Instructions

Question No. 19 #665df48dca324ed4a4c50dbe19 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

$f : R \to R$ is defined as $f(x) = \begin{bmatrix} x^2 + 2mx - 1 & \text{for } x \leq 0 \\ mx - 1 & \text{for } x > 0 \end{bmatrix}$

If $f(x)$ is one-one then m must lies in the interval

- $(-\infty, 0)$
- $(-\infty, 0]$
- $(0, \infty)$
- $[0, \infty)$

1 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20

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All Sections | Time Left: 174:32 | Question Paper | Instructions

Question No. 20 #665df48dca324ed4f4c50fbe20 | Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Let ΔABC is an isosceles triangle with $AB = AC$. If $B = (0, a)$, $C = (2a, 0)$ and the equation of AB is $3x - 4y + 4a = 0$, then the equation of side AC is

- $y = 8x - 16a$
- $3y = 4x - 8a$
- $x = 2a$
- $y + 8x = 16a$

0 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE CORRECT

Choose a question

1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
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All Sections | Time Left: 174:28 | Question Paper | Instructions

Question No. 1 #665df48dca324ed4a4c50fbe1 | Section Details | Marking Scheme

Mathematics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

If the solution for the differential equation $y^2dx + \left(x^2 - xy - y^2\right)dy = 0$ at $(2, 1)$ is $x + y = k\left(xy^2 - y^3\right)$, then $k =$

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4 Not Visited | 0 Marked for Review

0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS NUMERICAL

Choose a question

1 2 3 4 5

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All Sections | Time Left: 174:26 | Question Paper | Instructions

Question No. 2 #665df48dca324ed4a4d58fbe2 | Section Details | Marking Scheme

Mathematics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

$OABC$ is a unit square where O is the origin and $B = (1, 1)$. The curves $y^2 = x$ and $x^2 = y$ divide the area of the square into three parts $OABO$, OBO and $OBCO$. If a_1, a_2, a_3 are the areas (in sq units) of these parts respectively, then $a_1 + 2a_2 + 3a_3 =$

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Mark for Review & Next | Clear Response

3 Not Visited | 0 Marked for Review
0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS NUMERICAL

Choose a question

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All Sections | Time Left: 174:23 | Question Paper | Instructions

Question No. 3 #665df48dca324ed4a4d58fbe3 | Section Details | Marking Scheme

Mathematics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

A biased coin that has a probability of getting heads as p $(0 < p < 1)$ is tossed until a head appears for the first time. If the probability that the number of tosses required is even, is $\frac{2}{5}$, then $9p$ is equal to

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2 Not Visited | 0 Marked for Review
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MATHEMATICS NUMERICAL

Choose a question

1 2 3 4 5

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All Sections | Time Left: 174:20 | Question Paper | Instructions

Question No. 4 #665df48dca324ed4a4d5f0ee4 | Section Details | Marking Scheme

Mathematics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

If $x = \operatorname{cosec}\theta - \sin\theta, y = \operatorname{cosec}^{2022}\theta - \sin^{2022}\theta$ and $\left(\frac{dy}{dx}\right)^2 = \frac{k(y^2+4)}{g(x)}$ where $k \in \mathbb{R}$, then $10 + k - g(2022) =$

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4 5 6
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Left Right
Clear All

Mark for Review & Next | Clear Response | Save & Next | Submit

1 Not Visited | 0 Marked for Review
0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS NUMERICAL
Choose a question
1 2 3 4 5

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All Sections | Time Left: 174:17 | Question Paper | Instructions

Question No. 5 #665df48dca324ed4a4d58fbe5 | Section Details | Marking Scheme

Mathematics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

Number of skew-symmetric matrices of order 3 whose elements are $0, 0, 0, 1, -1, 2, -2, 3, -3$ is

Backspace
7 8 9
4 5 6
1 2 3
0 . -
Left Right
Clear All

Mark for Review & Next | Clear Response | Save & Next | Submit

0 Not Visited | 0 Marked for Review | 0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS NUMERICAL

Choose a question

1 2 3 4 5

>

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